

Problem statement

Current neuron-level explanation methods assume that specific neurons consistently represent stable semantic meaning. However, it is unclear whether these activations remain consistent when models process highly similar inputs. If neuron activations vary across similar statements, then explanations based on them may be unreliable and misleading.

Research Question

Do highly similar objective statements activate the same neurons in a neural network, or are neuron activations inconsistent and indistinguishable from random patterns?

Approach

This study tests neuron activation consistency using a pretrained transformer-based language model. Pairs of semantically similar but structurally varied objective statements are passed through the Bidirectional Encoder Representations from Transformers (BERT) natural language processing model. Neuron activations are recorded for each sentence in the pair for each layer using forward-pass hooks. The top 50 neurons based on activation values are then selected from each pair. The results from the similar sentences are then compared to random sentence pairs generated by randomly permuting the second sentences from the pairs. The similarity metrics for the similar pairs are then compared against the random pairs, which serve as the null distribution, using a significance test to determine whether neuron activation patterns are meaningfully consistent for each layer in the BERT model.

Dataset(s)

Stsb - <https://huggingface.co/datasets/sentence-transformers/stsb>

Dataset Description

The dataset utilized for this analysis consists of a variety of sentence pairs, formatted as strings, with an adjoined score as the third column. The first two columns each consist of a single sentence. The score column is a measure of the closeness or similarity of the second sentence to the first sentence, measured on a scale from 1 to 5 with 5 indicating that the sentences are identical in meaning, even if the sentences have different ordering or one or two synonyms. These scores were originally assigned by humans and have been scaled to be between 0 and 1. For our testing purposes, we selected all rows for which the score was greater than or equal to 0.8 to give us only sentence pairs that are considered to be highly similar. After removing rows that did not fit our similarity criteria, our dataset contains 264 rows of sentence pairs and scoring of the similarity in the sentiment between the two.

Metrics

The two metrics used to measure similarity of activation vectors are Jaccard and Cosine similarity. Jaccard similarity measures the proportion of shared neurons in the top-50 activated neurons for each pair, indicating overlap in which neurons have the highest activation values. Cosine similarity measures the similarity of the full activation vectors, capturing both which neurons are active and their relative magnitudes. Ultimately, the p-value of the significance test will determine whether we conclude that neuron activations are unreliable for similar objective statements